

Python Programming

Unit 06 – Lecture 06 Notes

Plot Types and Subplots

Tofik Ali

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1 Lecture Overview

Different plot types are used for different questions:

- trends → line plots,
- comparisons → bar charts,
- distributions → histograms,
- relationships → scatter plots.

Subplots help you combine multiple plots in one figure.

2 Core Concepts

2.1 Line Plot

Good for trends (time series).

```
plt.plot(x, y)
```

2.2 Bar Chart

Good for category comparisons.

```
plt.bar(categories, values)
```

2.3 Histogram

Good for distributions.

```
plt.hist(values, bins=10)
```

2.4 Scatter Plot

Good for relationships between two numeric variables.

```
plt.scatter(x, y)
```

2.5 Pie Chart (Use Carefully)

Pie charts show proportions, but can be hard to compare precisely.

```
plt.pie(values, labels=labels, autopct="%1.1f%%")
```

2.6 Subplots

Create multiple axes in one figure:

```
fig, ax = plt.subplots(2, 2, figsize=(8, 6))  
ax[0, 0].plot(x, y)
```

3 Demo Walkthrough

File: demo/matplotlib_plot_types_demo.py

The demo creates a 2x2 subplot figure and saves it into `images/`.

4 Interactive Checkpoints (with Solutions)

Checkpoint 1 Solution

Question: distribution of marks?

Answer: Histogram.

Checkpoint 2 Solution

Question: compare marks of 5 subjects?

Answer: Bar chart.

5 Practice Exercises (with Solutions)

Exercise 1: Histogram of Random Numbers

Solution:

```
import numpy as np
import matplotlib.pyplot as plt

x = np.random.randn(1000)
plt.hist(x, bins=20)
plt.title("Histogram")
plt.show()
```

6 Exit Question (with Solution)

Question: function name used to create subplots?

Answer: subplots

Appendix: Slide Deck Content (Reference)

The material below is a reference copy of the slide deck content. Exercise solutions are explained in the main notes where applicable.

Title Slide

Repository: <https://github.com/tali7c/Python-Programming>

Quick Links

[Core Concepts](#) [Demo](#) [Interactive](#) [Summary](#)

Agenda

- Core Concepts
- Demo
- Interactive
- Summary

Learning Outcomes

- Create different plot types: line, bar, histogram, scatter, pie
- Build multi-plot figures using subplots
- Choose the right plot type for the data story

Common Plot Types

- Line plot: trend over time
- Bar chart: compare categories
- Histogram: distribution of values
- Scatter plot: relationship between two variables
- Pie chart: parts of a whole (use carefully)

Subplots

- Subplots allow multiple plots in one figure
- Use `plt.subplots(rows, cols)`

```
fig, ax = plt.subplots(2, 2)
ax[0, 0].plot(x, y)
```

Demo: Multiple Plot Types in One Figure

- File: `demo/matplotlib_plot_types_demo.py`
- Generates a figure with subplots and saves it to `images/`

Checkpoint 1

Question: Which plot type would you use to show distribution of marks of 100 students?

Checkpoint 2

Question: Which plot type would you use to compare marks of 5 subjects?

Think-Pair-Share

Discuss:

- Why can pie charts be misleading?

Key Takeaways

- Different plot types answer different questions
- Subplots help compare multiple views of data in one figure
- Always label and choose scales carefully for honest visuals

Exit Question

Write the function name used to create subplots in Matplotlib.